

# Additional file 1

## The encoder, not the bins: decomposing peak-level MS/MS gains

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This file reports the per-arm and per-seed results behind every figure in the main text, the full bootstrap distributions, and the peak-absorption analysis broken down by acquisition condition. Every table here is also provided as a machine-readable CSV in the repository under `results/`, and every number is regenerated from saved run artefacts by `scripts/make_supplementary.py` and `scripts/make_si_figures.py`.

### S1 Per-arm, per-seed metrics

Table S1 gives every trained arm on both corpora. Twenty-four training runs: four arms  $\times$  three seeds  $\times$  two corpora. Parameter counts are identical between M1R and M1 by construction, and the binned baseline M0 carries 63% more parameters than either.

**Table S1:** Held-out metrics for every arm and seed. Macro AUPRC is the primary metric; chance is 0.0085 on the open corpus and 0.0093 on the single-library corpus. Machine-readable: `results/si_table.s1_arm_metrics.csv`.

| corpus         | arm | seed | parameters | best epoch | macro AUPRC | macro AUROC | Brier   |
|----------------|-----|------|------------|------------|-------------|-------------|---------|
| open           | M0  | 2026 | 1,164,560  | 14         | 0.1120      | 0.8607      | 0.00555 |
| open           | M0  | 7    | 1,164,560  | 14         | 0.1143      | 0.8613      | 0.00554 |
| open           | M0  | 13   | 1,164,560  | 13         | 0.1126      | 0.8610      | 0.00554 |
| open           | M1R | 2026 | 712,336    | 14         | 0.1591      | 0.8991      | 0.00525 |
| open           | M1R | 7    | 712,336    | 13         | 0.1570      | 0.8980      | 0.00525 |
| open           | M1R | 13   | 712,336    | 14         | 0.1541      | 0.8960      | 0.00526 |
| open           | M1  | 2026 | 712,336    | 14         | 0.1731      | 0.9063      | 0.00513 |
| open           | M1  | 7    | 712,336    | 13         | 0.1709      | 0.9063      | 0.00517 |
| open           | M1  | 13   | 712,336    | 13         | 0.1687      | 0.9068      | 0.00514 |
| open           | M2  | 2026 | 829,584    | 14         | 0.1899      | 0.9072      | 0.00512 |
| open           | M2  | 7    | 829,584    | 14         | 0.1893      | 0.9052      | 0.00511 |
| open           | M2  | 13   | 829,584    | 12         | 0.1871      | 0.9090      | 0.00512 |
| single library | M0  | 2026 | 1,164,560  | 13         | 0.0476      | 0.7832      | 0.00704 |
| single library | M0  | 7    | 1,164,560  | 13         | 0.0472      | 0.7819      | 0.00704 |
| single library | M0  | 13   | 1,164,560  | 14         | 0.0485      | 0.7837      | 0.00704 |
| single library | M1R | 2026 | 712,336    | 13         | 0.0867      | 0.8457      | 0.00673 |
| single library | M1R | 7    | 712,336    | 14         | 0.0856      | 0.8474      | 0.00672 |
| single library | M1R | 13   | 712,336    | 14         | 0.0840      | 0.8470      | 0.00675 |
| single library | M1  | 2026 | 712,336    | 14         | 0.1119      | 0.8734      | 0.00654 |
| single library | M1  | 7    | 712,336    | 13         | 0.1085      | 0.8717      | 0.00656 |
| single library | M1  | 13   | 712,336    | 13         | 0.1162      | 0.8759      | 0.00652 |
| single library | M2  | 2026 | 829,584    | 13         | 0.1460      | 0.8861      | 0.00640 |
| single library | M2  | 7    | 829,584    | 13         | 0.1428      | 0.8824      | 0.00644 |
| single library | M2  | 13   | 829,584    | 14         | 0.1444      | 0.8837      | 0.00641 |

## S2 Per-seed effect sizes and ordering checks

Each effect is the difference between two arms trained under the *same* seed, so the three values per row are matched rather than independent samples (Table S2). Where the main text claims an ordering between two effects, it is checked within each seed; Table S3 reports those checks.

**Table S2:** Effect sizes per seed, as differences in held-out macro AUPRC.

| corpus         | effect                | seed 2026 | seed 7  | seed 13 | mean    | SD     |
|----------------|-----------------------|-----------|---------|---------|---------|--------|
| open           | encoder (M1R – M0)    | +0.0471   | +0.0427 | +0.0415 | +0.0438 | 0.0030 |
| open           | mass axis (M1 – M1R)  | +0.0140   | +0.0139 | +0.0147 | +0.0142 | 0.0004 |
| open           | aggregation (M2 – M1) | +0.0168   | +0.0184 | +0.0184 | +0.0179 | 0.0009 |
| single library | encoder               | +0.0392   | +0.0383 | +0.0355 | +0.0377 | 0.0019 |
| single library | mass axis             | +0.0252   | +0.0229 | +0.0322 | +0.0268 | 0.0049 |
| single library | aggregation           | +0.0341   | +0.0343 | +0.0282 | +0.0322 | 0.0035 |

**Table S3:** Ordering checks. Each entry is the within-seed difference between two effects; an ordering is reported in the main text only where it holds in all three seeds.

| corpus         | comparison              | seed 2026 | seed 7  | seed 13 | mean    | verdict         |
|----------------|-------------------------|-----------|---------|---------|---------|-----------------|
| open           | encoder – mass axis     | +0.0331   | +0.0288 | +0.0268 | +0.0296 | 3/3 consistent  |
| open           | encoder – aggregation   | +0.0303   | +0.0243 | +0.0231 | +0.0259 | 3/3 consistent  |
| open           | mass axis – aggregation | –0.0028   | –0.0045 | –0.0037 | –0.0037 | 3/3 consistent  |
| single library | encoder – mass axis     | +0.0139   | +0.0154 | +0.0033 | +0.0109 | 3/3 consistent  |
| single library | encoder – aggregation   | +0.0051   | +0.0040 | +0.0074 | +0.0055 | 3/3 consistent  |
| single library | mass axis – aggregation | –0.0089   | –0.0114 | +0.0041 | –0.0054 | <b>reverses</b> |

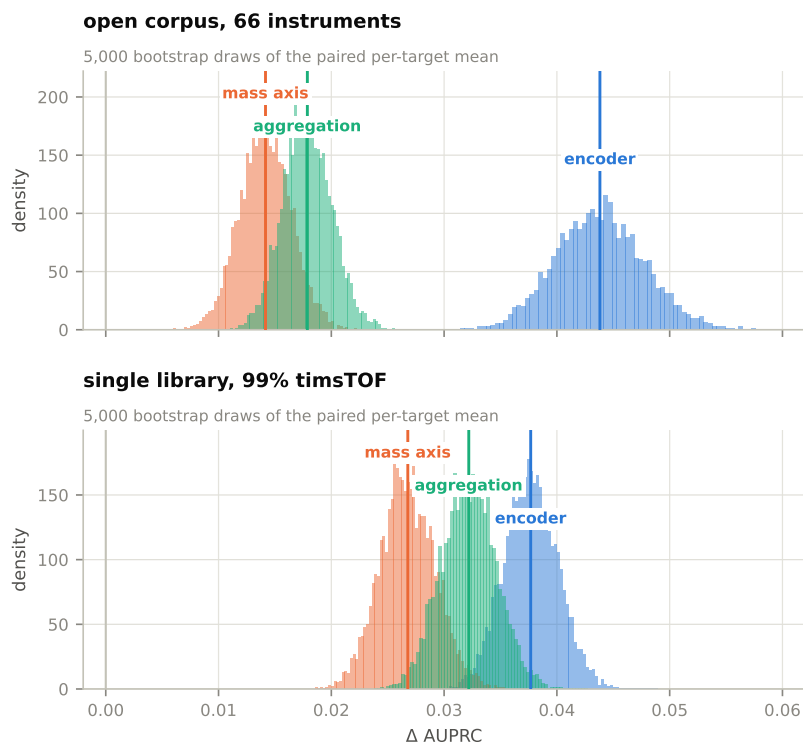
The final row is why the main text ranks the mass axis against aggregation on the open corpus and declines to on the single-library corpus.

## S3 Bootstrap distributions

Figure S1 shows the full distribution behind every interval reported in the main text: 5,000 resamples of the mean paired per-target difference, over targets, after averaging each target’s average precision across the three seeds. Table S4 summarises them.

**Table S4:** Bootstrap summaries. Machine-readable draws: `results/si_table.s3_bootstrap.draws.csv.gz`.

| corpus         | effect      | mean    | SD      | 2.5%    | 97.5%   |
|----------------|-------------|---------|---------|---------|---------|
| open           | encoder     | 0.04381 | 0.00391 | 0.03637 | 0.05173 |
| open           | mass axis   | 0.01416 | 0.00229 | 0.00972 | 0.01876 |
| open           | aggregation | 0.01788 | 0.00218 | 0.01372 | 0.02223 |
| single library | encoder     | 0.03768 | 0.00239 | 0.03311 | 0.04245 |
| single library | mass axis   | 0.02678 | 0.00243 | 0.02208 | 0.03168 |
| single library | aggregation | 0.03219 | 0.00243 | 0.02751 | 0.03696 |



**Figure S1:** Bootstrap distributions of the three effects on each corpus. On the open corpus the encoder distribution is separated from the other two; on the single-library corpus all three overlap, and the mass-axis and aggregation distributions overlap heavily — which is the same fact Table S3 records as a sign reversal.

## What these intervals do and do not cover

Each interval resamples targets, having first averaged average precision over the three seeds. It therefore describes uncertainty from the finite set of targets, not from initialisation. Seed variability is reported separately as the SD column of Table S2, and the two should be read together: on the single-library corpus the mass-axis effect has a seed SD of 0.0049, comparable to the width of its bootstrap interval, which is the statistical form of the ordering failure in Table S3. A variance-components treatment with targets and seeds as crossed random effects would fold both into one interval and we regard it as the right analysis for a larger replicate count.

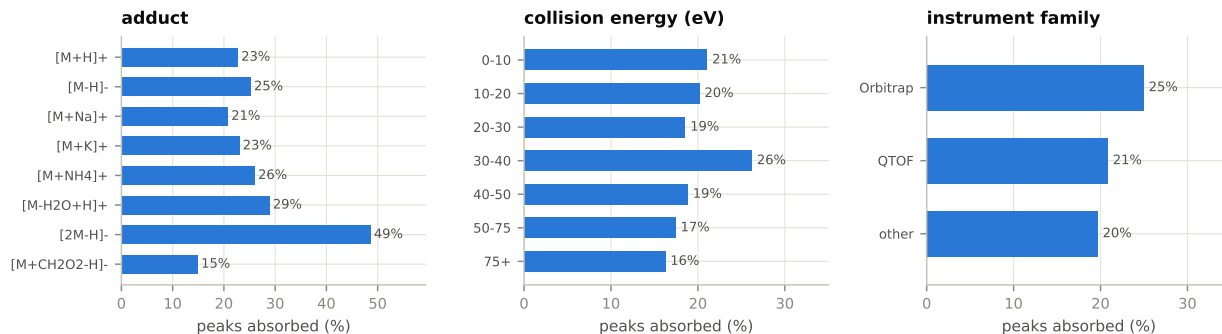
## S4 Per-target average precision

`results/si_table_s2_per_target_ap.csv` gives average precision for every scored target, for every arm, both as the seed-averaged value used in the paired analysis and per seed: 433 targets on the open corpus and 477 on the single-library corpus, 910 rows in total. All paired differences in the main text are recoverable from this file by subtraction.

## S5 Absorption by acquisition condition

The main text reports that 23.6% of the open corpus’s held-out peaks fall into a 0.5 Da bin already occupied. Figure S2 breaks that down by adduct, collision energy and instrument family.

**Absorption at 0.5 Da by acquisition stratum, open corpus held-out spectra**



**Figure S2:** Absorption at 0.5 Da by acquisition stratum, over the open corpus’s held-out spectra. Strata with fewer than 20,000 peaks are omitted.

Absorption tracks peak density rather than any chemical property of the adduct. The highest rate among common adducts is  $[2M-H]^-$  at 48.7%, whose spectra are dimeric and correspondingly crowded;  $[M+H]^+$ , which supplies 62% of held-out spectra, sits at 22.8%. Across collision energies the range is 16% to 26% with no monotone trend. Orbitrap spectra lose 24.9% against QTOF’s 20.8%, again following density (52.6 against 47.2 peaks per spectrum) rather than resolving power.

This is worth stating because it constrains the interpretation in the main text. Binning costs more where peaks are dense, and density varies with acquisition conditions that have nothing to do with how much mass precision the instrument actually delivered.

Machine-readable, all under `results/`:

`si_absorption_by_adduct.csv`  
`si_absorption_by_collision_energy.csv`  
`si_absorption_by_instrument_family.csv`

## S6 Bin-width sweep

Table S5 gives the sweep in full. Every arm uses the M1 encoder at an identical parameter count; only the width of the mass grid changes. One run per width, on the single-library corpus.

**Table S5:** Bin-width sweep, single seed, single-library corpus.

| bin width (Da) | peaks absorbed | macro AUPRC | share of the 0.5 Da gap closed |
|----------------|----------------|-------------|--------------------------------|
| 0.5            | 37.5%          | 0.0867      | 0%                             |
| 0.25           | 36.8%          | 0.0818      | −19%                           |
| 0.1            | 33.1%          | 0.0801      | −26%                           |
| 0.05           | 25.9%          | 0.0909      | 17%                            |
| 0.01           | 7.5%           | 0.1030      | 65%                            |
| full precision | 0.0%           | 0.1119      | 100%                           |

The absorption column here is computed over the competition test spectra, which are denser than the corpus’s own held-out spectra (72.5 against 58 peaks per spectrum); the corresponding held-out figures are 34.4% at 0.5 Da falling to 6.5% at 0.01 Da, and those are the numbers the main text uses.

## S7 Reproduction

Software: Python 3.10, PyTorch 2.11.0+cu128, RDKit 2026.03.6, NumPy 2.4.4, pandas 2.3.3, pyarrow 24.0.0. Exact pins are in `requirements.txt`.

The full path from the corpus file to every number in this file is

```
python scripts/make_open_subset.py --root .
python -m dbf2 prepare --root . --dataset open
python -m dbf2 ablate --root . --dataset open \
    --rungs M0 M1R M1 M2 --epochs 15
python -m dbf2 ablate --root . --dataset open \
    --rungs M0 M1R M1 M2 --epochs 15 --seed 7
python -m dbf2 ablate --root . --dataset open \
    --rungs M0 M1R M1 M2 --epochs 15 --seed 13
python scripts/make_supplementary.py
python scripts/make_si_figures.py
```

with the same three commands minus `--dataset open` for the single-library corpus, and

```
for w in 0.25 0.1 0.05 0.01; do
    python -m dbf2 train --root . --rung M1 \
        --quantise $w --tag W$w --epochs 15
done
```

for the sweep. `python -m pytest src/dbf2/tests` runs 50 unit tests, 14 of which cover the rounding control, including that M1R and M1 have identical parameter counts and that their configurations differ in exactly one field.

#### **A note on what the subset builder reads**

`make_open_subset.py` selects the GNPS, MassBank and MoNA rows out of the aggregated CASMI 2026 file. Those spectra are openly licensed at their sources, but reproducing this work as written still requires that aggregated file. Fetching the records from GNPS, MassBank and MoNA directly is recorded as an open item in the repository. `docs/corpus_composition.md` reports each source library's contribution and states which parts are redistributable.